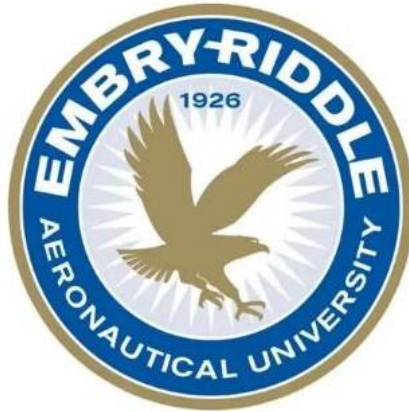


Flexible Link Modeling



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Procedure

A. Finding Stiffness

In the first part of the lab, the stiffness of the flexible link is calculated from the measured natural frequency. Before experimenting, it is important to confirm that all the lab files are configured in the system. The link was modeled as a second-order system, and its stiffness was calculated using the relationship between natural frequency and stiffness. This experiment was carried out by first manually oscillating the flexible link while keeping the servo base fixed. Using the given Simulink model, the strain gauge sensor was used to record the oscillatory response. The period of oscillation was calculated by plotting the link angle's time history data that was recorded to the MATLAB workspace. After calculating the natural frequency from the measured period, the link stiffness was calculated using the relationship, $K = J\omega_n^2$.

B. Model Validation

The goal of the second part of the lab is to verify if the dynamic model is accurate by comparing simulated results to the measured results of the actual device. The experiment starts by running the “setup_flexgage.m” script. When prompted, enter the stiffness found in the previous evaluation. This also enables the state-space matrices A, B, C, and D to be loaded in the workspace. Since the physical system is driven by motor voltage rather than torque, actuator dynamics were incorporated into the model by modifying the A and B matrices according to the given script in the manual.

The q_flexgage_val Simulink model was then configured with the appropriate solver settings and hardware configuration. Upon running the simulation, the servo angle (θ) and flexible link angle (α) were measured from the hardware while the simulated responses were generated in parallel. Both answers were automatically stored to the MATLAB workspace. The amplitude, oscillation frequency, damping behavior, and transient features of the simulated and observed responses were compared. In order to examine system stability and dynamic behavior, the state-space matrices were also used to generate the open-loop poles and transfer function. This step confirmed whether the mathematical model adequately represented the physical flexible link system.

Results

Elastic Potential Energy:

$$V = \frac{1}{2} K_s \alpha^2$$

Kinetic Energy:

$$T = \frac{1}{2} J_{eq} \dot{\theta}^2 + \frac{1}{2} J_l (\dot{\theta} + \dot{\alpha})^2$$

Lagrangian:

$$L = T - V$$

Forces:

$$Q_1 = \tau - B_{eq} \dot{\theta}$$

$$Q_2 = -B_l \dot{\alpha}$$

Euler-Lagrange:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = Q_1$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\alpha}} \right) - \frac{\partial L}{\partial \alpha} = Q_2$$

$$\frac{\partial L}{\partial \dot{\theta}} = (J_{eq} + J_l) \dot{\theta} + J_l \dot{\alpha}$$

$$\frac{\partial L}{\partial \dot{\alpha}} = J_l (\dot{\theta} + \dot{\alpha})$$

$$\frac{\partial L}{\partial \alpha} = \frac{\partial}{\partial \alpha} \left(-\frac{1}{2} K_s \alpha^2 \right)$$

Servo Equation:

$$(J_{eq} + J_l) \ddot{\theta} + J_l \ddot{\alpha} = \tau - B_{eq} \dot{\theta}$$

Link Equation:

$$J_l \ddot{\theta} + J_l \ddot{\alpha} + B_l \dot{\alpha} + K_s \alpha = 0$$

Solving both equations:

$$\ddot{\theta} = \frac{1}{J_{eq}} (\tau - B_{eq} \dot{\theta} + B_l \dot{\alpha} + K_s \alpha)$$

$$\ddot{\alpha} = -\frac{1}{J_{eq}} \tau + \frac{B_{eq}}{J_{eq}} \dot{\theta} - \left(\frac{B_l}{J_{eq}} + \frac{B_l}{J_l} \right) \dot{\alpha} - \left(\frac{K_s}{J_{eq}} + \frac{K_s}{J_l} \right) \alpha$$

State Matrices:

$$x = \begin{bmatrix} \theta \\ \alpha \\ \dot{\theta} \\ \dot{\alpha} \end{bmatrix}, u = \tau$$

$$\dot{x}_1 = x_3$$

$$\dot{x}_2 = x_4$$

$$\dot{x}_3 = \ddot{\theta} = \frac{K_s}{J_{eq}} x_2 - \frac{B_{eq}}{J_{eq}} x_3 + \frac{B_l}{J_{eq}} x_4 + \frac{1}{J_{eq}} u$$

$$\dot{x}_4 = \ddot{\alpha} = -\left(\frac{K_s}{J_{eq}} + \frac{K_s}{J_l} \right) x_2 + \frac{B_{eq}}{J_{eq}} x_3 - \left(\frac{B_l}{J_{eq}} + \frac{B_l}{J_l} \right) x_4 - \frac{1}{J_{eq}} u$$

Assuming, $B_l = 0$

Hence,

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{K_s}{J_{eq}} & -\frac{B_{eq}}{J_{eq}} & 0 \\ 0 & -\left(\frac{K_s}{J_{eq}} + \frac{K_s}{J_l} \right) & \frac{B_{eq}}{J_{eq}} & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{J_{eq}} \\ -\frac{1}{J_{eq}} \end{bmatrix}$$

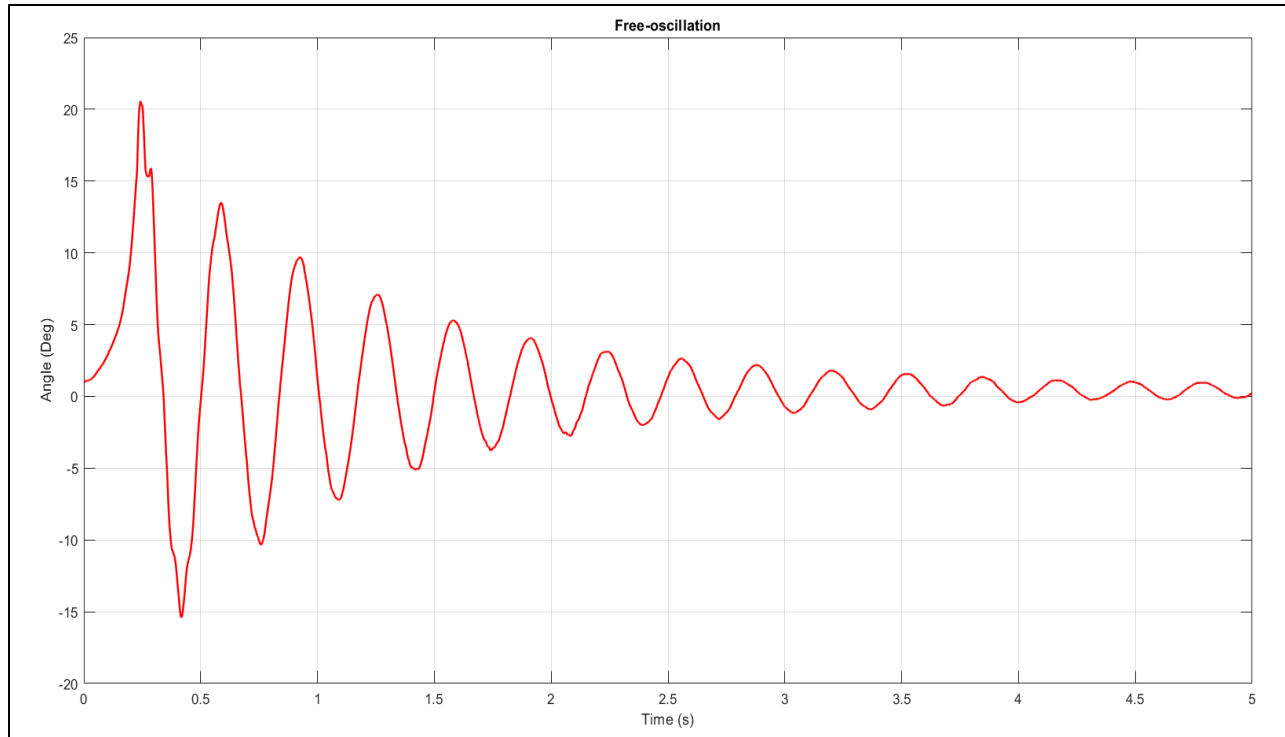


Figure 1: Free-Oscillation Plot

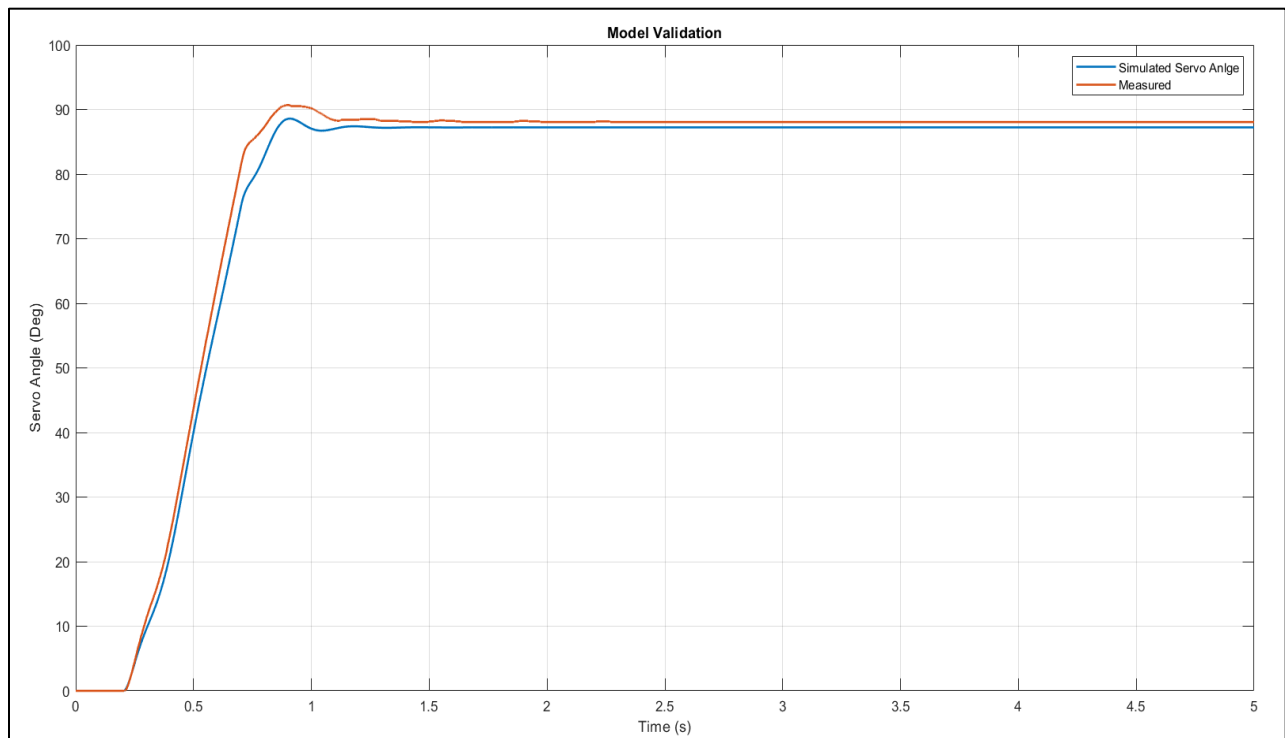


Figure 2: Model Validation Plot – Servo Angle

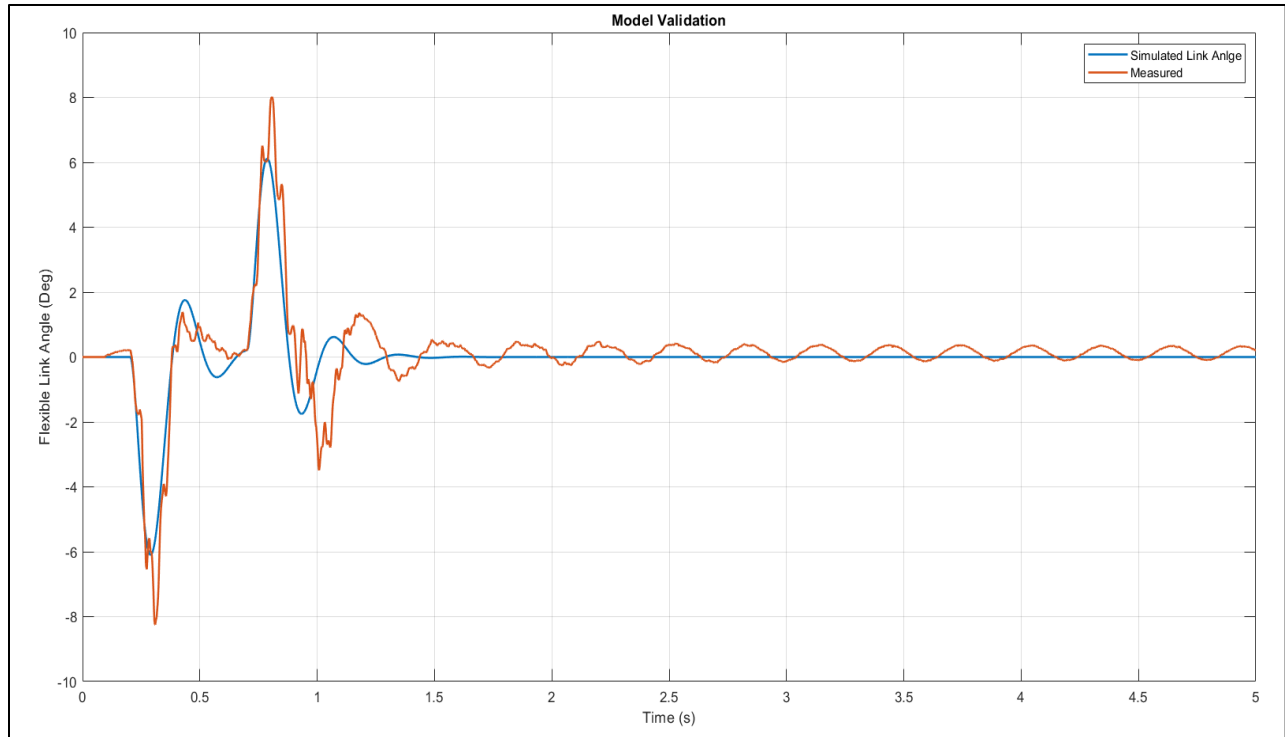


Figure 3: Model Validation Plot – Flexible Link Angle

Table 1: Modeling Lab Results

Description	Symbol	Value	Unit
Finding Stiffness			
Natural Frequency	ω_n	18.49	rad/s
Stiffness	K_s	1.30	(N · m)/rad
Model Validation			
System Matrix	A	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 623.7468 & -40.3205 & 0 \\ 0 & -965.3452 & 40.3205 & 0 \end{bmatrix}$	
Control Input Matrix	B	$\begin{bmatrix} 0 \\ 0 \\ 61.6326 \\ -61.6326 \end{bmatrix}$	
Output Matrix	C	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$	

Feed-Forward Matrix	D	$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$	
Transfer Function	-	$\frac{61.63s^2 + 2.105 \times 10^4}{s^4 + 40.32s^3 + 965.3s^2 + 1.377 \times 10^4s}$ $\frac{-61.63s + 6.193 \times 10^{-13}}{s^3 + 40.32s^2 + 965.3s + 1.377 \times 10^4}$	
Open-loop Poles	-	$P_1 = 0$ $P_{2,3} = -8.1565 \pm 22.5207i$ $P_4 = -24.0076$	

Analysis

Natural Frequency Calculations:

The first step is to select the set of reference points from the Free-Oscillation plot, which will be used to calculate various parameters. Figure 4 shows the selected points from the plot where MATLAB was used to determine the data by using the cursor.

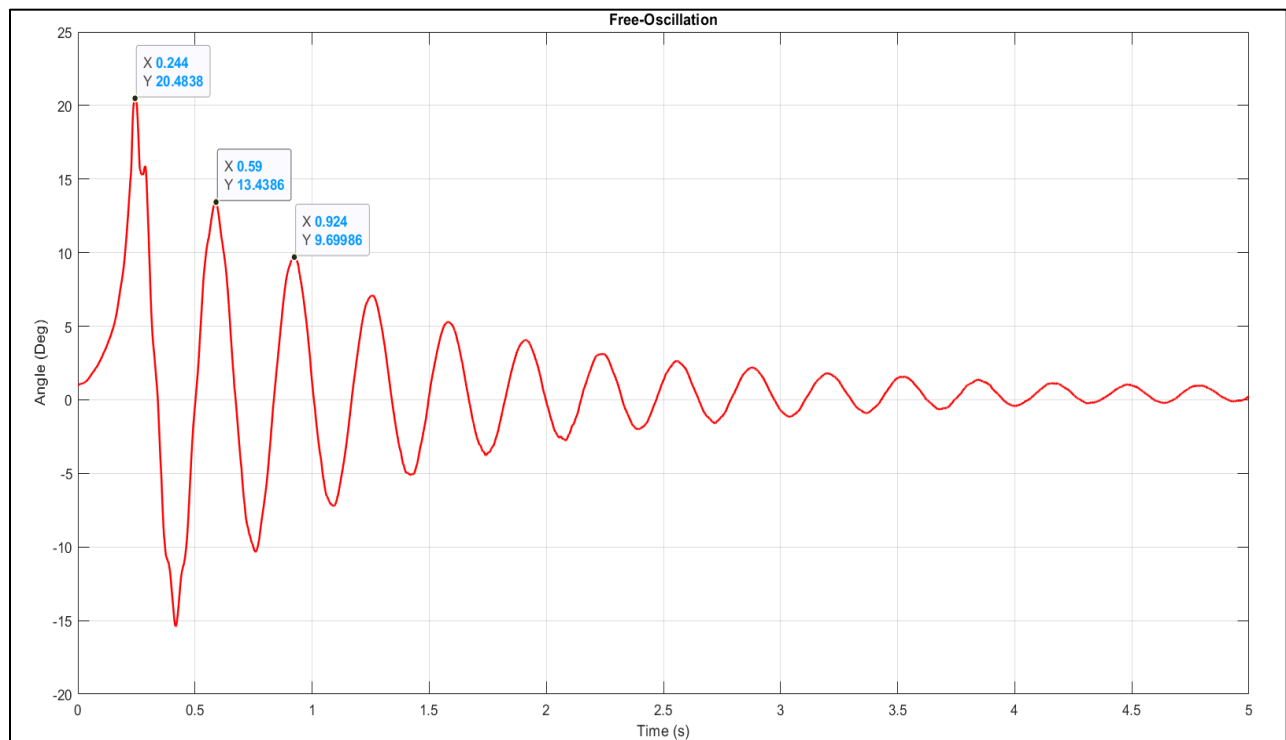


Figure 4: Free-Oscillation plot with Data Points

Table 2: Data Points

n	Time (s)	Angle (deg)	Symbol
1	0.244	20.48	O_1
2	0.592	13.38	O_2
3	0.924	9.70	O_3

Next, the following equations are used, which are provided in the lab manual, in the same chronological order.

Period of oscillations:

$$T_{osc} = \frac{t_{n+1} - t_1}{n} = \frac{0.924 - 0.244}{2} = 0.34 \text{ s}$$

Damped natural frequency:

$$\omega_d = \frac{2\pi}{T_{osc}} = \frac{2\pi}{0.34} = 18.48 \frac{\text{rad}}{\text{s}}$$

Subsidence ratio (decrement ratio):

$$\delta = \frac{1}{n} \ln \frac{O_1}{O_n} = \frac{1}{2} \ln \frac{20.48}{13.38} = 0.213$$

Damping ratio:

$$\zeta = \frac{1}{\sqrt{1 + \left(\frac{2\pi}{\delta}\right)^2}} = \frac{1}{\sqrt{1 + \left(\frac{2\pi}{0.213}\right)^2}} = 0.0339$$

Natural Frequency:

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} = \frac{18.48}{\sqrt{1 - 0.0339^2}} = 18.49 \frac{\text{rad}}{\text{s}}$$

[Link Stiffness Calculations:](#)

Before calculating the stiffness, some physical parameters of the lab are required to be recorded, which are provided in the lab manual. Table 3 illustrates these values.

Table 3: Lab Parameters

Parameter	Value
B_{eq}	$0.004 \text{ N} \cdot \text{m}/(\text{rad/s})$
J_{eq}	$2.08 \times 10^{-3} \text{ kg} \cdot \text{m}^2$
m_l	0.065 kg
L_l	0.419 m

Next, the following equations are used to determine the link stiffness.

Moment of Inertia of link:

$$J_l = \frac{m_l \cdot L_l^2}{3} = \frac{0.065 \times 0.419^2}{3} = 0.0038 \text{ kg} \cdot \text{m}^2$$

Stiffness of flexible link:

$$K = J\omega_n^2 = 0.0038 \times 18.49^2 = 1.30 \frac{\text{N} \cdot \text{m}}{\text{rad}}$$

Characteristic equation, open-loop poles, transfer function:

The finished state-space matrices A, B, C, and D were used to directly derive the characteristic equation, open-loop poles, and transfer function. In MATLAB, the system's transfer function representation was created by first creating the state-space model and then transforming it into transfer function form (see Appendix A). Because the output vector includes both the flexible link angle (α) and the servo angle (θ), the resulting transfer function is a 2×1 system that shows $\frac{\theta(s)}{u(s)}$ and $\frac{\alpha(s)}{u(s)}$ with respect to the input u . By calculating the eigenvalues of the system matrix A, the open-loop poles of the system were identified. These poles characterize the open-loop system's natural modes and correspond to the roots of the characteristic equation $\det(sI - A) = 0$.

The model not representing the system accurately can be explained by the following reasons. First, simplifying assumptions such as viscous damping and linear spring behavior are used to build the state-space model. The mathematical model does not account for nonlinear effects that are present in the flexible link and servo system in practice, such as changes in friction, minor structural nonlinearities, and faulty sensors. Second, theoretical calculations are utilized to determine the model's parameters, which include stiffness, inertia and damping coefficients. The accuracy of the matrices A and B is immediately impacted by any error in detecting the natural frequency or calculating the stiffness, which results in differences between the simulated and measured responses. Furthermore, the model ignores some hardware issues like amplifier limitations, measurement noise, and

slight delays in the data acquisition system, and assumes perfect actuator dynamics. During model validation, these unmodeled dynamics help explain the discrepancies between the simulated and real responses.

Conclusion

The objectives of this experiment were successfully completed. By measuring the flexible link's natural frequency from the free-oscillation response, the stiffness of the link was determined in the first lab section. The natural frequency was derived from experimental data, and the link was modeled as a second-order system with light damping. Using the calculated moment of inertia and the relationship $K = J\omega_n^2$, the stiffness of the link was found to be 1.30 N·m/rad. This demonstrated that the physical properties of a mechanical system may be derived from experimental data.

The generated state-space model was verified in the second part of the experiment by comparing measured hardware responses with simulated responses. The transfer function was produced using the state-space representation, and the open-loop poles were derived from the eigenvalues of the system matrix A. One pole was at zero, whereas the other poles had negative real portions. This shows that the pole at zero causes the system to be marginally stable, whereas the complex conjugate pair with negative real parts causes the flexible dynamics to be stable and oscillatory.

The simulated and measured responses displayed comparable trends in rising time, oscillation frequency, and overall transient behavior. However, slight variations in damping and amplitude were noted. This suggests that the model does not accurately depict the physical system, even though it captures the primary features of the flexible link system. In general, the state-space model offers a fair representation of the behavior of the actual system. The experiment showed how the dynamics of a flexible mechanical system can be analyzed and validated by combining state-space modeling and experimental identification.

References

- [1] Moncayo H. Ph.D. Student Workbook Experiment 02. Embry-Riddle Aeronautical University. 2023.
- [2] Quanser Inc. QUARC User Manual.
- [3] Quanser Inc. SRV02 QUARC Integration, 2008.
- [4] Quanser Inc. QUARC Installation Guide, 2009.

[5] Quanser Inc. SRV02 User Manual, 2009.

[6] Quanser Inc. QUARC Compatibility Table, 2010.

[7] Quanser Inc. SRV02 Rotary Flexible Link User Manual, 2011.

[8] Norman S. Nise. Control Systems Engineering. John Wiley & Sons, Inc., 2008.

Appendix A

MATLAB Code:

```
% Author: Anuranan Bharadwaj
% AE 443 - Lab 2

%% Exp 1

figure();
plot(data_alpha(:,1), data_alpha(:,2), 'r', 'LineWidth', 1.5);
grid on;
xlabel("Time (s)");
ylabel("Angle (Deg)");
title("Free-Oscillation");

%% Exp 2 - a

figure();
plot(data_theta(:,1), data_theta(:,2), 'LineWidth', 1.5);
hold on
plot(data_theta(:,1), data_theta(:,3), 'LineWidth', 1.5);
grid on;
xlabel("Time (s)");
ylabel("Servo Angle (Deg)");
title("Model Validation");
legend('Simulated Servo Angle', 'Measured')

%% Exp 2 - b

figure();
plot(data_alpha(:,1), data_alpha(:,2), 'LineWidth', 1.5);
hold on
plot(data_alpha(:,1), data_alpha(:,3), 'LineWidth', 1.5);
grid on;
xlabel("Time (s)");
ylabel("Flexible Link Angle (Deg)");
title("Model Validation");
legend('Simulated Link Angle', 'Measured')

%%
sys = ss(A, B, C, D);
TF = tf(sys)

poles = eig(A)
```